



MCS3301 – Control Engineering I

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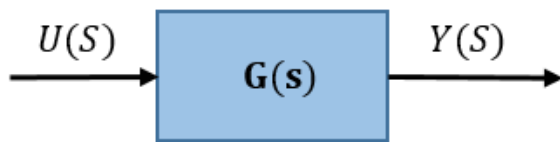
Block Diagram Models

- Block diagrams provide control engineers with a better understanding of the **composition** and **interconnection** of the components of a system.
- Block diagrams consist of **unidirectional**, operational blocks that represent the **transfer function** of the **variables of interest**.
- They can be used together with transfer functions, to describe the cause-and-effect relationships throughout the system.
- The common elements in block diagrams of most control systems include
 - ✓ Comparators
 - ✓ Blocks representing individual component transfer functions, including: sensors, controllers, plants and actuators
 - ✓ Input or reference signals
 - ✓ Output signals
 - ✓ Disturbance signals
 - ✓ Feedback loops

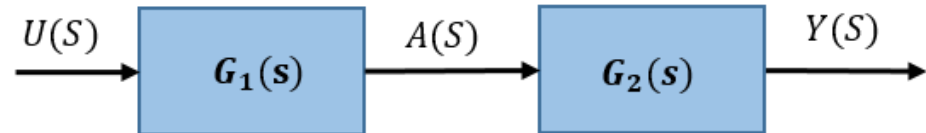


Block Diagram Models

- **Blocks:** Blocks represent the equations describing the block elements which are typically **plants**, **controllers**, **actuators** or **sensors**.



$$Y(s) = G(s)U(s)$$



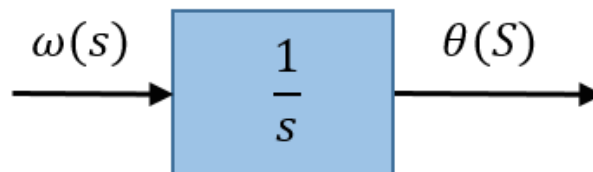
$$Y(s) = A(s)G_2(s)$$

$$A(s) = U(s)G_1(s)$$

$$Y(s) = G_1(s)G_2(s)U(s)$$

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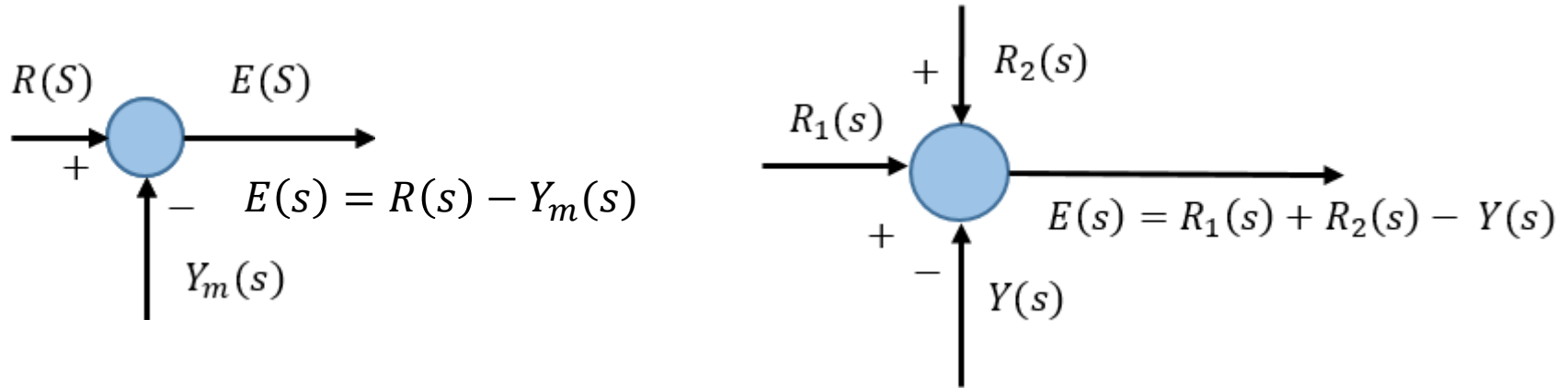
- Blocks are also commonly used to represent **integrators** in systems. For example the output of block below $\theta(s)$ (position) is the integral of the **angular speed** input $\omega(s)$.



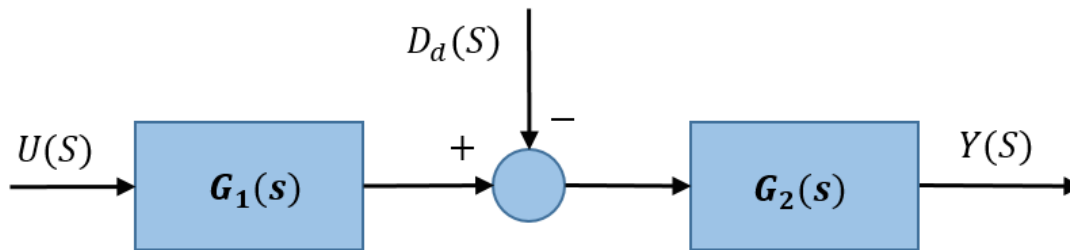


Block Diagram Models

- **Comparators:** These are junction points for **signal comparisons**. They perform linear operations of addition and subtraction.



- These junctions are also used to model **disturbance** signals to a system as shown below

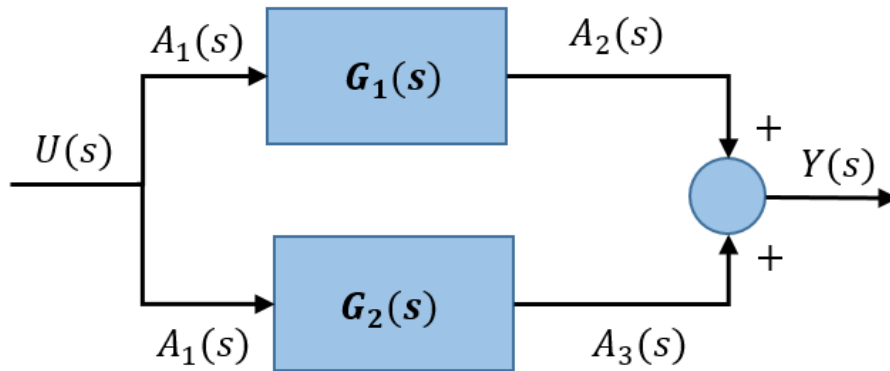


$$Y(s) = G_2(G_1 U(s) - D_d(s))$$



Block Diagram: Example

Find the transfer function of the system shown below



$$A_2 = A_1(s)G_1(s)$$

$$A_3 = A_1(s)G_2(s)$$

$$Y(s) = A_2(s) + A_3(s)$$

$$Y(s) = U(s)(G_1(s) + G_2(s))$$

$$G(s) = \frac{Y(s)}{U(s)} = G_1(s) + G_2(s)$$



Block Diagram From Mathematical Eqns.

- Consider the general expression for a second-order system

$$\frac{d^2x(t)}{dt^2} + 2\zeta\omega_n \frac{dx(t)}{dt} + \omega_n^2 x(t) = \omega_n^2 u(t)$$

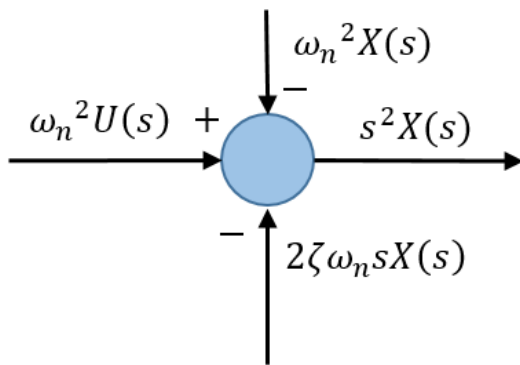
- Its Laplace transform, assuming zero initial conditions is given by

$$s^2X(s) + 2\zeta\omega_n sX(s) + \omega_n^2 X(s) = \omega_n^2 U(s)$$

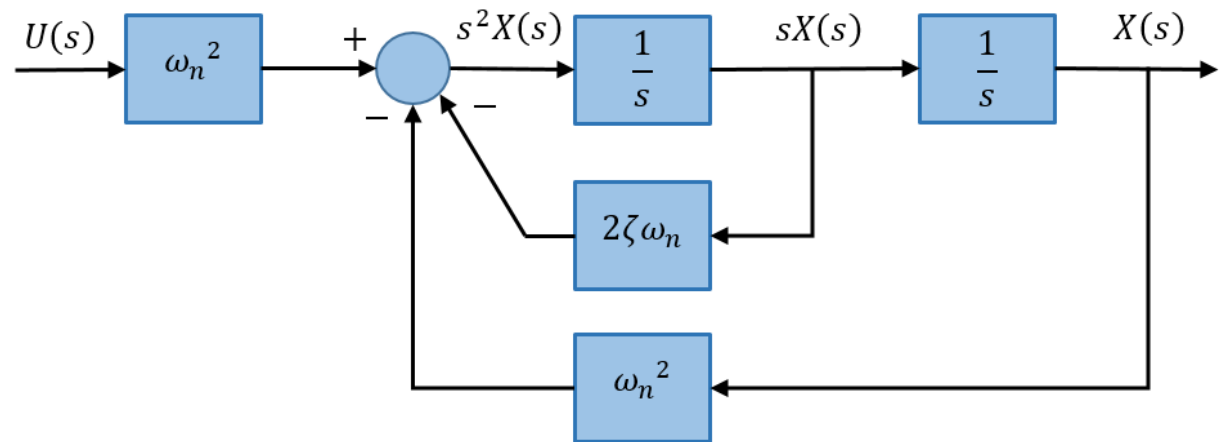
- Rearranging gives

$$\omega_n^2 U(s) - 2\zeta\omega_n sX(s) - \omega_n^2 X(s) = s^2X(s)$$

- This can be represented graphically using a comparator or block diagrams as shown below.



Graphical representation
using a comparator



Block diagram representation



Block Diagram Manipulation

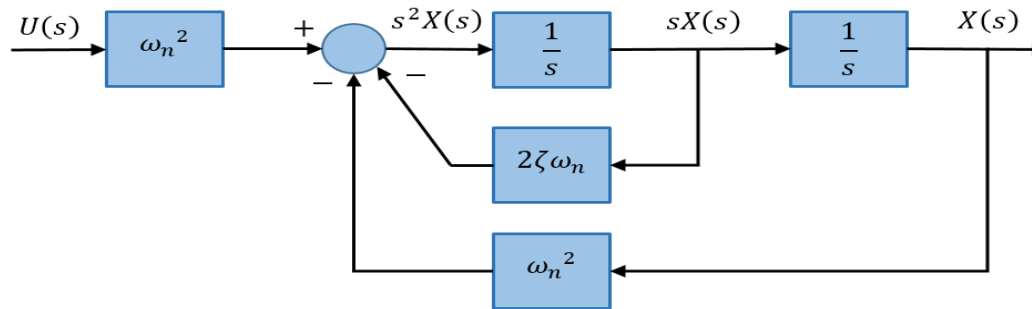
- Some of the commonly used rules in manipulating block diagrams are listed in the table below

S/N	Transformation	Equation	Block Diagram	Equivalent Block Diagram
1	Combining cascaded blocks	$Y = (G_1 G_2)X$		
2	Eliminating a forward loop	$Y = G_1 X \pm G_2 X$		
3	Eliminating a feedback loop	$Y = G_1 (X \pm G_2 Y)$		
4	Moving a summing point ahead of a block	$Z = GX \pm Y$		
5	Moving a summing point beyond a block	$Z = G(X \pm Y)$		

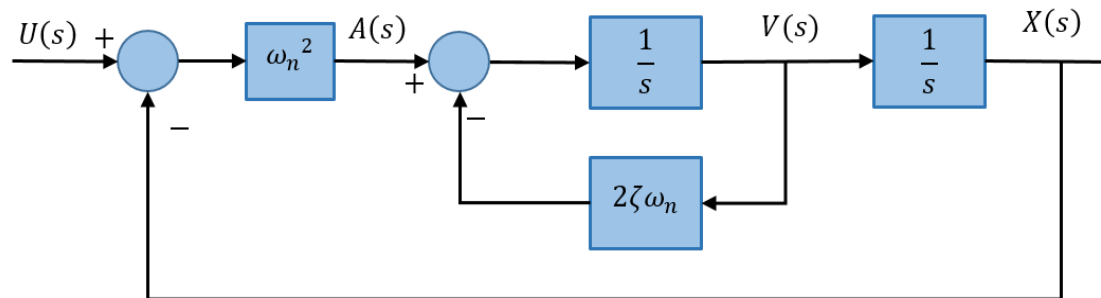


Block Diagram Manipulation: Example

- Find an expression for the transfer function of the system below by reducing the block diagram to a single block.



- The ω_n^2 block at the **input** and **feedback** can be moved to the right hand side of the summing (comparator) block.



- This is the same as factorization by ω_n^2 and the two block diagrams shown are equivalent.

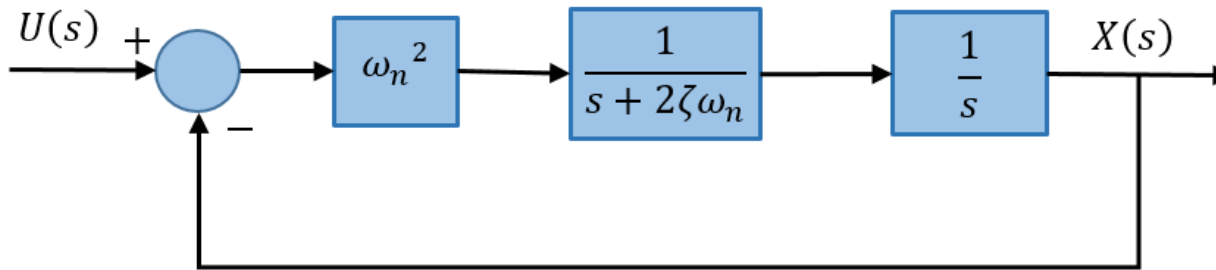
$$\omega_n^2 U(s) - \omega_n^2 X(s) = \omega_n^2 (U(s) - X(s))$$



Block Diagram Manipulation: Example

- The **inner feedback** loop can next be eliminated

$$\frac{V(s)}{A(s)} = \frac{1/s}{1 + \frac{1}{s} \cdot 2\zeta\omega_n} = \frac{1}{s + 2\zeta\omega_n s}$$



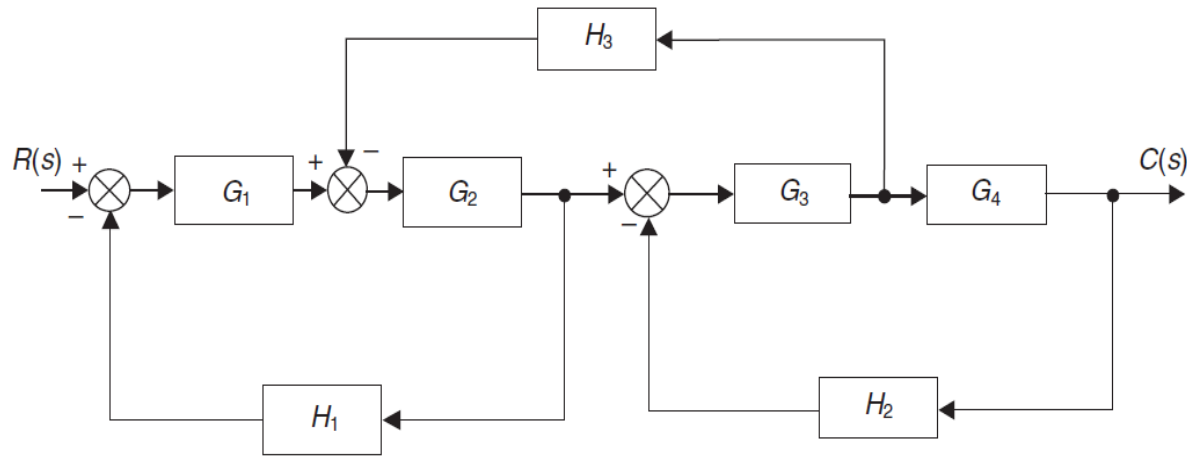
- Eliminate the **outer feedback** loop to give

$$\frac{X(s)}{U(s)} = \frac{\frac{\omega_n^2}{s(s + 2\zeta\omega_n)}}{1 + \frac{\omega_n^2}{s(s + 2\zeta\omega_n)}} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

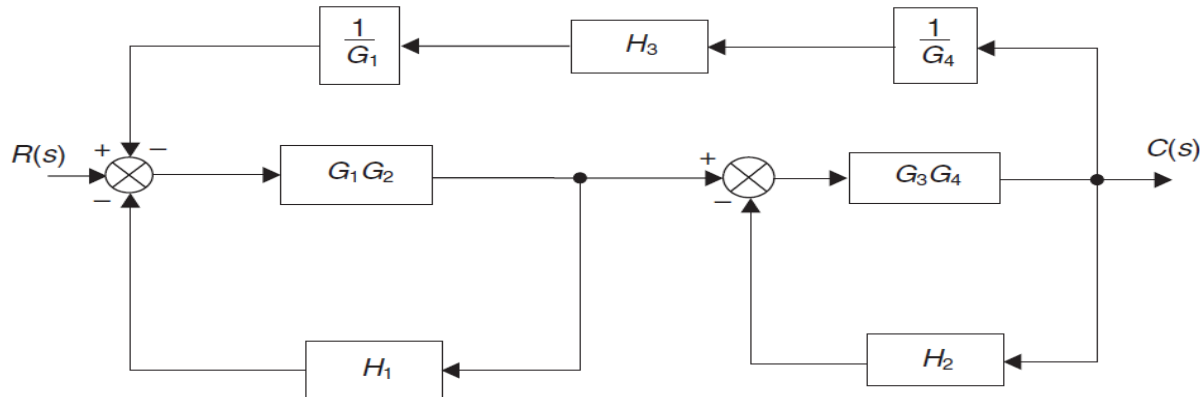


Block Diagram Reduction Example

- Find the overall closed loop transfer function of the system below via block diagram reduction.



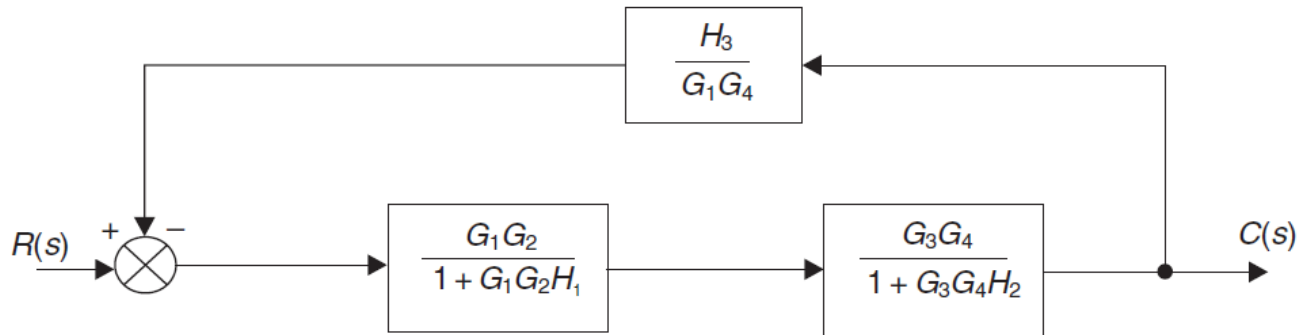
- Move the H_3 feedback loop **ahead** of the G_1 block and **beyond** the G_4 block.





Block Diagram Reduction Example

- Eliminate the H_1 and H_2 feedback loops.



- Finally the overall transfer function can be expressed as

$$\frac{C(s)}{R(s)} = \frac{\frac{G_1 G_2 G_3 G_4}{(1 + G_1 G_2 H_1)(1 + G_3 G_4 H_2)}}{1 + \frac{G_1 G_2 G_3 G_4}{(G_1 G_4)(1 + G_1 G_2 H_1)(1 + G_3 G_4 H_2)}}$$

$$\frac{C(s)}{R(s)} = \frac{G_1 G_2 G_3 G_4}{(1 + G_1 G_2 H_1)(1 + G_3 G_4 H_2) + G_2 G_3 H_3}$$